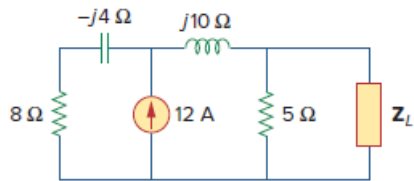


Problem

For the circuit shown in Fig. 11.10, find the load impedance $\mathbf{Z}_L$ that absorbs the maximum average power. Calculate that maximum average power.

Figure 11.10:



Step-by-step solution

Step 1 of 9

Consider the circuit in Figure 1.

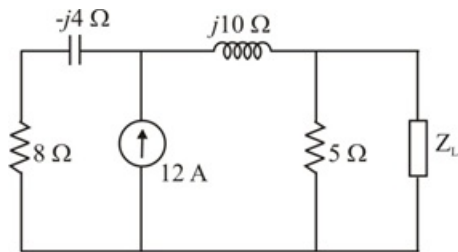


Figure 1

Step 2 of 9

Find the Thevenin's impedance $\mathbf{Z}_{Th}$.

Consider the following circuit to be considering that the current source is kept open as shown in Figure 2.

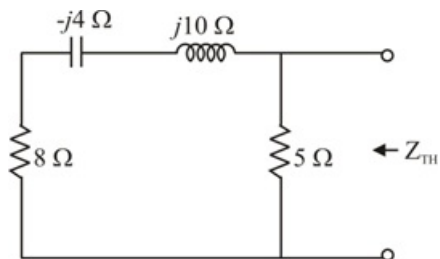


Figure 2

Step 3 of 9

Calculate the Thevenin's equivalent impedance $\mathbf{Z_{Th}}$ across the load terminals:

$$\begin{aligned}\mathbf{Z_{Th}} &= (8\ \Omega - j4\ \Omega + j10\ \Omega) \parallel (5\ \Omega) \\ &= (8 + j6) \parallel (5)\ \Omega \\ &= \frac{(8 + j6)(5)}{8 + j6 + 5}\ \Omega \\ &= (3.414 + j0.73)\ \Omega\end{aligned}$$

Step 4 of 9

The load impedance $\mathbf{Z_L}$ absorbs the maximum average power if,

$$\begin{aligned}\mathbf{Z_L} &= \mathbf{Z_{Th}^*} \\ &= (3.414 + j0.731)^* \ \Omega \\ &= (3.414 - j0.731)\ \Omega\end{aligned}$$

Thus, the load impedance $\mathbf{Z_L}$ that absorbs the maximum average power is

$$\boxed{(3.414 - j0.731)\ \Omega}.$$

Here,

$$\mathbf{Z_L} = R_L + jX_L\ \Omega$$

Hence,

$$R_L = 3.414\ \Omega$$

$$X_L = 0.73\ \Omega$$

Step 5 of 9

Find Thevenin's equivalent voltage $\mathbf{V_{Th}}$.

Consider the circuit as shown in Figure 3.

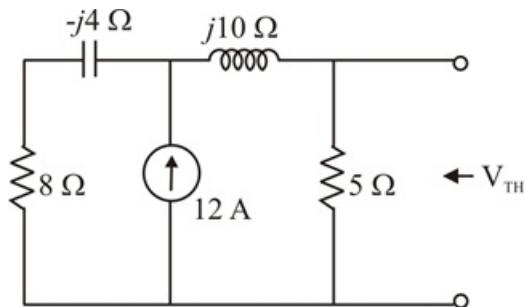


Figure 3

Step 6 of 9

Convert the current source to voltage source $\mathbf{V_s}$.

$$\begin{aligned}\mathbf{V_s} &= \mathbf{I_s Z} \\ &= (12\ \text{A})(8\ \Omega - j4\ \Omega) \\ &= (12)(8 - j4) \\ &= 96 - j48 \\ &= 107.33 \angle -26.56^\circ\ \text{V}\end{aligned}$$

Step 7 of 9

Redraw the circuit as shown in Figure 4.

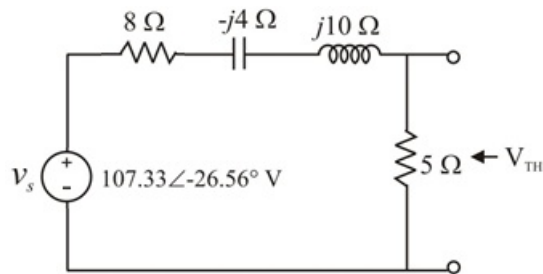


Figure 4

Step 8 of 9

Calculate the value of V_{Th} .

$$\begin{aligned} V_{Th} &= \frac{5 \Omega}{(8 - j4 + j10 + 5) \Omega} (96 - j48 \text{ V}) \\ &= \frac{5(96 - j48)}{(13 + j6)} \\ &= \frac{5(96 - j48)}{(13 + j6)} \\ &= (23.414 - j29.268) \text{ V} \quad \text{or} \\ &= 37.48 \angle -51.34^\circ \text{ V} \end{aligned}$$

Hence, the maximum voltage across the load is,

$$V_M = 37.48 \text{ V}$$

Step 9 of 9

Calculate the value of maximum average power P_{max} .

$$\begin{aligned} P_{max} &= \frac{V_{rms}^2}{4R_L} \\ &= \frac{\left| \frac{V_M}{\sqrt{2}} \right|^2}{4R_L} \\ &= \frac{|V_M|^2}{8R_L} \end{aligned}$$

Substitute 37.48 V for V_M and 3.414 Ω for R_{Th} .

$$\begin{aligned} P_{max} &= \frac{(37.48 \text{ V})^2}{8(3.414 \Omega)} \\ &= \frac{(37.48)^2}{8(3.414)} \\ &= 51.43 \text{ W} \end{aligned}$$

Thus, the value of the maximum average power is 51.43 W.